Does the Fed Move Tesla? A Time-Series Analysis of Treasury Yields and TSLA Stock Returns

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1. Dataset Description

This project analyzes the relationship between Tesla Inc. (TSLA) stock prices and U.S. 10-Year Treasury Yields over the period January 2020 to December 2024. Two datasets were used. The first is TSLA daily closing price data sourced from Yahoo Finance via the yfinance Python library. The second is the 10-Year Treasury Constant Maturity Rate (DGS10) sourced from the Federal Reserve Economic Data (FRED) database via the FRED API. The two datasets were merged on matching trading dates using an inner join, resulting in a final clean dataset of 1,158 observations.

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2. Data Cleaning & Preprocessing

Both datasets were inspected for data quality issues prior to analysis. No missing values, duplicate dates, or negative prices were found in either raw dataset. After merging, daily returns were computed for both TSLA prices and Treasury yields using percentage change. A MinMaxScaler was applied to normalize both the TSLA price and 10-Year yield series to a 0–1 scale, enabling direct comparison between the two series. The single NaN introduced by the percentage change calculation was dropped, and regime and volatility columns were computed, yielding a final clean dataset of 1,158 rows across 7 columns.

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3. Exploratory Data Analysis

Six visualizations were produced to explore patterns and relationships in the data.

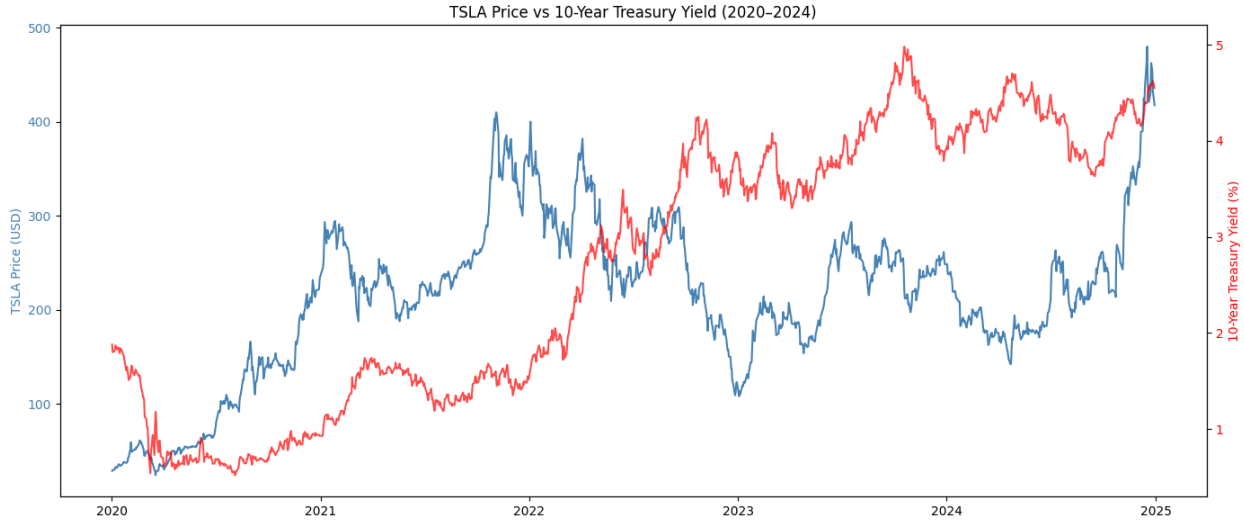

Chart 1 — TSLA Price vs 10Y Treasury Yield: The dual-axis chart reveals a clear inverse relationship between TSLA price and Treasury yields over the five-year period. During 2020–2021, yields remained near historic lows of 0.5–1% while TSLA surged from approximately $30 to $400. As the Federal Reserve began aggressively hiking rates in 2022, yields climbed toward 4–5% and TSLA collapsed by over 70% from its peak. Through 2023–2024, yields remained elevated and TSLA struggled to recover before a late 2024 rally despite persistently high yields — a notable anomaly.

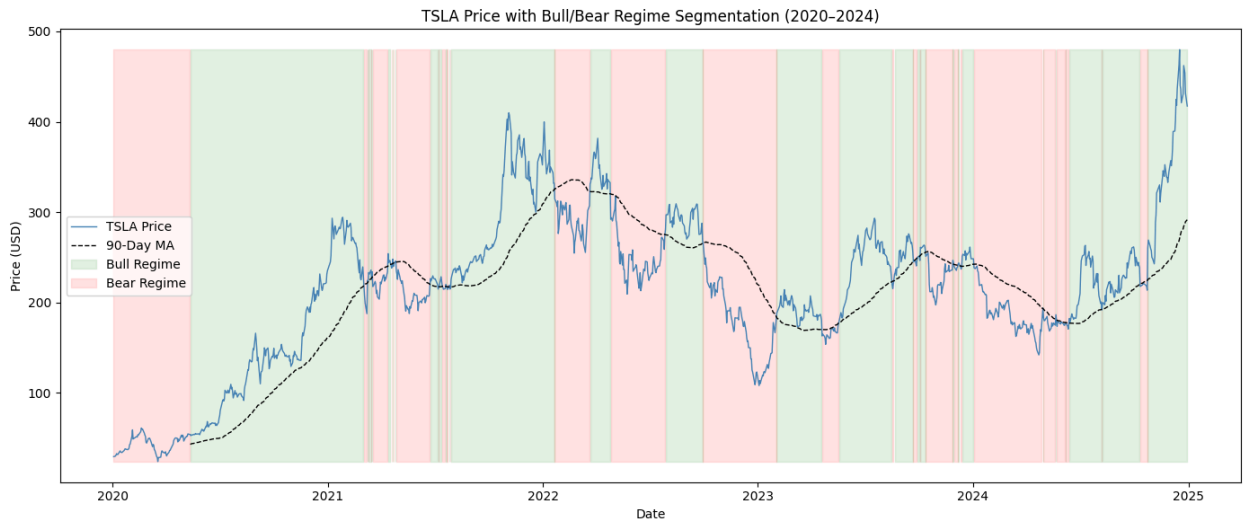

Chart 2 — Bull/Bear Regime Segmentation: Using a 90-day moving average as a threshold, TSLA's price history was segmented into bull and bear regimes. The chart clearly identifies the COVID crash bear period in early 2020, the extended bull regime through 2020–2021, the bear regime during the 2022 rate hike cycle, and the choppy transition period of 2023 before the late 2024 bull recovery. The frequent regime flipping in 2023 indicates an indecisive, range-bound market.

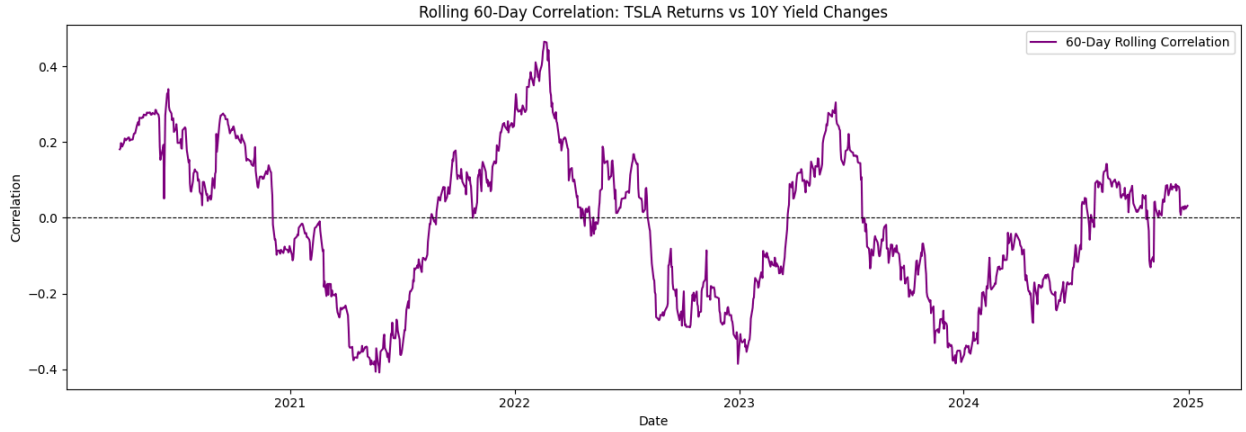

Chart 3 — Rolling 60-Day Correlation: The rolling correlation between TSLA returns and yield changes oscillated between +0.4 and -0.4 over the period. The most strongly negative readings occurred during the peak rate hike cycle of 2022–2023, confirming that the inverse relationship strengthened precisely when monetary policy became the dominant market narrative.

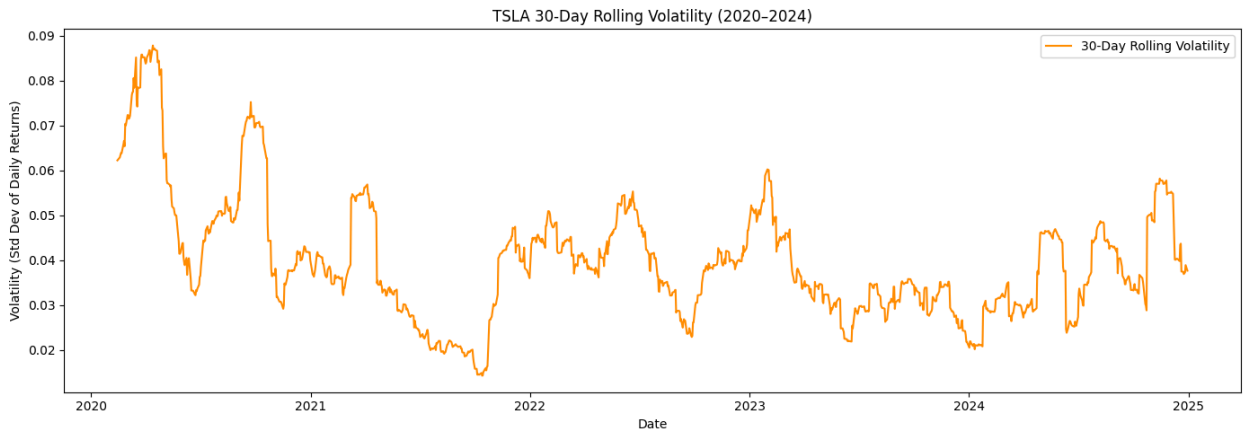

Chart 4 — Rolling 30-Day Volatility: TSLA exhibited its highest volatility during the COVID crash of early 2020, with a secondary spike during its explosive 2020–2021 rally. Volatility remained elevated through the 2022 rate hike selloff before moderating in 2023–2024, though TSLA remained an inherently volatile stock throughout the entire period.

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4. Statistical Modeling

Granger Causality Test: A Granger causality test was applied to determine whether past changes in Treasury yields contain statistically significant predictive information about future TSLA returns. Results were statistically significant at lags 1, 2, 3, and 5 (p-values of 0.0084, 0.0193, 0.0387, and 0.0149 respectively), confirming at the 95% confidence level that Treasury yield movements Granger-cause TSLA returns.

Vector Autoregression (VAR) Model: A VAR model was fitted to the joint series of TSLA returns and yield changes. The optimal lag order of 4 was selected automatically using the Akaike Information Criterion (AIC = -13.64). The model was evaluated using RMSE (0.0383) and MAE (0.0328), indicating an average daily forecast error of approximately 3.3–3.8% — reasonable given TSLA's inherent volatility.

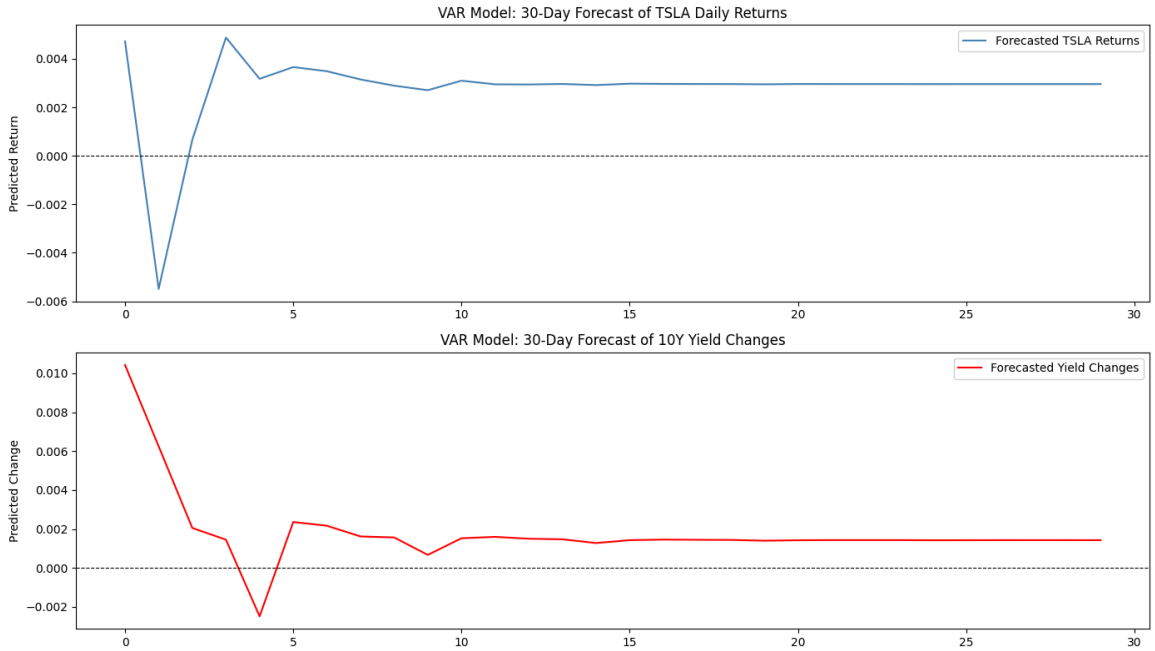

Chart 5 — VAR 30-Day Forecast: The VAR model forecast modest positive TSLA daily returns over the subsequent 30-day period, coinciding with stabilizing yield changes near zero — consistent with a plateauing rate environment heading into 2025.

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5. Key Findings & Insights

Three principal findings emerged from this analysis:

First, Treasury yields and TSLA prices exhibit a clear and statistically significant inverse relationship. When yields rise, TSLA tends to fall, and when yields fall, TSLA tends to rally. This is consistent with the theory that rising rates compress the present value of future earnings, disproportionately impacting high-growth stocks like Tesla.

Second, the relationship is not static — it strengthens during periods of aggressive monetary policy and weakens when company-specific factors dominate. This was most evident in the late 2024 rally, where TSLA surged despite persistently high yields, suggesting that non-rate factors temporarily overrode the macroeconomic relationship.

Third, the Granger causality test provides statistical confirmation that yield movements contain genuine predictive information about TSLA's near-term direction, particularly at 1–3 day lags.

From a strategy perspective, monitoring Federal Reserve policy signals and Treasury yield movements may provide useful leading indicators for positioning in growth stocks like TSLA.

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6. Reflection

This project demonstrated a statistically significant and visually compelling relationship between U.S. Treasury yields and Tesla stock returns. The combination of regime segmentation, rolling correlation, Granger causality testing, and VAR modeling provided a multi-layered view of the relationship that goes beyond simple price observation.

Key limitations include the fact that Granger causality does not imply true economic causation, and that TSLA is an unusually idiosyncratic stock influenced by many factors beyond interest rates. Future work could incorporate additional variables such as the VIX volatility index, TSLA earnings data, or broader market returns to build a more complete predictive model.